1 Hypothesis

What I have: Improvements on Prototype 1 and Prototype 2 and Prototype 4 Whats the difference between them: Prototype 1 and 2 are handmade maps while Prototype 4 is procedurally generated. What is similar : The map assets, the horror aesthetic , the mechanics. Will the addition of mechanics that allow the player to traverse the map in different ways allow the use of bigger maps.

2 Goal

- Make a story.
- Connect all the prototypes.
- Improve the prototype algorithm.

3 Story

The player will find themselves in a sort of laboratory where a disembodied voice will try to guide the player out of the laboratory. If the player follows the voice they will successfully escape the laboratory. If the player disobeys the voice they will be taken deeper into the laboratory where they will then have to escape from the laboratory without the help of the narrator.

4 Game Progression

The general game progression is that the player will start in the tutorial and then make their ways from the first level to the third level. The tutorial and first level will be handmade and will have a narrator on both levels. The second and third level will both be procedurally generated with different values in their generation to indicate a decay of the laboratory.

4.1 Tutorial

The player will start in the tutorial where they will learn the basics of the game. The player will first learn the basic controls which are: Moving, sprinting, toggling the player light and general interactions the player can perform. The player will be introduced to the enemy in the tutorial and different ways to avoid them.

4.2 First Level

The player will be put in a laboratory map where a disembodied voice called the narrator will then try to help the player escape from the laboratory. If the player follows the narrator then they will escape from the laboratory, if the player disobeys the narrator they will go deeper into the laboratory. The enemy will not appear in this level. This level will not be procedurally generated.

4.3 Second Level

The player will be in a laboratory that is starting to decay. The player will then need to complete a puzzle while hiding and avoiding the enemy. After the player completes the puzzle they will be taken to the next level through either a door opening on the map or being teleported there. The decay will be shown by giving some random walk to the procedural generation.

4.4 Third Level

The player will be in a laboratory that is decayed. The player will then face the same challenges that they faced in the second level. The decay is shown by giving the procedural generation a high level of random walk.

5 Genre

The genre of the game will be a stealth survival horror and puzzle genre. The player will need to hide and evade an unstoppable enemy while trying to complete some sort of goal. This will be similar to the Alien from Alien Isolation, because the AI enemy was inspired from Alien Isolation. The puzzle genre appears when the player will need to find solutions to the puzzles presented to them.

6 Possible Puzzles

These are the puzzles that can appear during the procedural generation of the level. Each puzzle will have success criteria which is determined by the puzzle.

6.1 Fetch quest

The player will need to find an item and bring it to the destination.
The generation will be a success if:

- The item can be accessed.
- The item can be picked up and put down.
- The destination can be accessed.
- The destination recognizes the item.
- The player completes the goal after brining the item to the destination.

6.2 Password

The player will need to find a password that they will then use to open a lock. The generation will be a success if:

- All parts of the password can be accessed
- All parts of the password can be read.
- The lock can be accessed.
- The lock successfully recognizes the entered password.
- The lock completes the goal after the password has been entered.

6.3 Pattern

The player will be presented with a special room in which the player will need to move objects in a specific order to complete the puzzle. The generation will be a success if:

- All parts of the puzzle can be accessed
- All parts of the puzzle can be picked up and put down.
- The puzzle can be accessed.
- The puzzle successfully recognizes the entered pattern.
- The puzzle completes the goal after the pattern has been entered.

7 Prototype Improvements

This prototype will improve upon all previous prototypes, while connecting all the prototypes together.

7.1 Prototype 1 Improvements

The map of prototype 1 will be improved while also allowing the player to use more of the mechanics to explore the map. This will include areas that can only be accessed when the player's light is turned off and areas that have a timer that the player needs to sprint towards to access. This will give the player more interaction in exploring the map.

7.2 Prototype 2 Improvements

The map of prototype 2 will be improved to make the tutorial a more optional experience instead of a forced experience. It will gradually show the player the mechanics without overwhelming them by showing them all the mechanics at once. A small tutorial will be shown to the player in the pause ui so the player can be reminded about the controls at any point during the game.

7.3 Prototype 3 Improvements

This prototype focused on the enemy thus the enemy will be improved. The enemy will be given a static state in which they will not move. In this static state the enemy will only listen for player footsteps and move towards them. The enemy sight will also be shown in the tutorial to the player in this state to help the prototype 2 improvements. This new state can get the player to possible stun the enemy allowing for ways to fight against but not remove the enemy.

7.4 Prototype 4 Improvements

The procedural generation algorithm will be improved by keeping track of all the rooms in the generation to allow adding of special rooms which will either be a room in which a puzzle needs to be solved or will have an interaction for the player. The algorithm will also allow for secret passages between rooms.

8 Communication Design

9 Process

9.1 Tutorial Design

The design philosophy for the tutorial is that it needs to introduce the player to the general mechanics. This means it first needs to show the player how to move and then after confirming that the player knows how to move will then

9.2 Prototype 1

9.3 Prototype 2

9.4 Prototype 3

Adding enemy teleporters for the enemy to go to specific areas. The map made for prototype 3 was just to show off the enemy thus it has been removed making the final level progression of the game be: Prototype 2, Prototype 1, Prototype 4.

9.5 Prototype 4

If enemy out of the map teleport the enemy back to starting position. Increased the player light size because the original light size made the player feel too cramped in the room causing the player to get lost in the map a bit too often. The original size was a 4 unit circle while the new size is a 7 unit circle. The detection for the enemy to see the player light is still the original size to allow the player some leeway in not getting spotted by the enemy but still being able to see what is happening.

A minimap was added to help the player navigate, but this minimap took away from the game by giving the player a sense of direction. This takes away from the horror elements of the game because the player feels a sense of dread because they are lost, which the minimap diminishes. The current procedural generation generates the map in a maze-like way which may be too confusing for players, thus the minimap helps the players traverse the map. Both improvements want to be kept thus a solution was found. The solution is that if the player uses the minimap the enemy will be alerted to their position and will constantly go to the position of the player. This solution makes the minimap a limited resource that the player can use at a risk. This leads to the horror elements of the game staying by letting the player feel like they are in danger because of the use of the minimap, but it also helps the player traverse the maze-like map reducing the player's frustration. The minimap will ping the players position every few seconds which will guide the enemy to the player. Text was added to the puzzle solution to help the player remember which parts of the solution the player has already found. This was done because the players would usually forget the solution parts they have found because the enemy chasing them causes the players to forget what they just found in order to try escape the enemy.

10 references (delete subsections later)

10.1 Tutorials

https://medium.com/@doandaniel/gamedev-protips-a-few-helpful-tips-on-crafting-a-better-tutorial-ffcc636039d3 https://www.gamedeveloper.com/business/gamedev-thoughts-how-to-design-a-good-tutorial-for-your-indie-game

10.2 Minimap

https://pixabay.com/sound-effects/film-special-effects-radar-ping-306440/